Power BI Session 2

Introduction to DAX — Measures — Calculated Columns — Advanced Analytics

Dataset: Business data
Duration: 2-3 hours — **Level:** Intermediate

EXERCISE 1: MEASURES VS CALCULATED COLUMNS

Objective: Understand the difference between measures and calculated columns in DAX.

Task:

Create both a calculated column and a measure, then analyze their differences.

Questions:

a) **Create a calculated column named Revenue Calculation with formula:**

```
= [Quantity] * [Unit_Price]
```

Now answer:

- What does this calculated column represent? the revenu
- Does every row have a value for this column? yes
- What is the value for the first row? 1465.11

b) **Create a measure named Total Revenue Measure with formula:**

```
= SUM([Revenue])
```

Now answer:

- What does this measure calculate? the total revenu
- Can you put this measure in a table and see individual row values? no we can't see it in a table because it's only one value
- What is the total value of this measure? 1.24M

c) **Compare the calculated column and measure:**

- Which one evaluates for each row individually? the calculated column
- Which one changes based on the current filter context? the calculated measure

- When would you use each one? if you want to see how every row changes individually you need to choose the calculated column.

d) **When is it better to use a calculated column instead of a measure?**

EXERCISE 2: CREATING BASIC MEASURES

Objective: Create fundamental measures for business metrics using aggregation functions.

Task:

Create the following measures using basic DAX aggregation functions:

Measure 1: Total Profit

```
= SUM([Profit])
```

a) What is the Total Profit value? 606.28K

b) Filter by Product Category = "Electronics". What is the Total Profit now? 129.21K

c) How does the measure value change when you apply a filter? it dicreases

Measure 2: Average Order Value

```
= AVERAGE([Revenue])
```

2.23K

Measure 3: Total Order Count

```
= COUNTROWS(Business dataset)
```

500

EXERCISE 3: CALCULATED COLUMNS

Objective: Create calculated columns to derive new insights from existing data.

Task:

Create the following calculated columns:

Column 1: Profit Margin Percentage

```
= DIVIDE([Profit], [Revenue], 0) * 100
```

a) What is the profit margin for the first order? 56.94%

b) What is the profit margin for the order with highest revenue? 59.76%

c) Find an order with negative profit margin. What does this indicate? -42 there is a deficit

d) What is the average profit margin across all orders? 50.03%

Column 2: Revenue Category

```
= IF([Revenue] >= 5000, "High",
   IF([Revenue] >= 2000, "Medium", "Low"))
```

e) How many orders are in each Revenue Category? (Count for each)

- High: 49
- Medium: 166
- Low: 285

f) What percentage of orders are in the "High" revenue category?

g) Does the High revenue category represent high profit as well? Explain.

Column 3: Discount Applied Flag

```
= IF([Discount_Percent] > 0, "Yes", "No")
```

h) How many orders have discounts applied? 396

i) What percentage of orders include a discount? 79.2%

j) Do discounted orders have lower profit margins? Create a simple analysis.

EXERCISE 4: FILTER CONTEXT

Objective: Create measures that are aware of filter context using CALCULATE.

Task:

Create measures that respond to filters in different ways:

Measure 1: Revenue for Selected Product Category

```
= CALCULATE(SUM([Revenue]),[Product_Category] ="Electronics")
```

Measure 2: Revenue for All Categories (Ignoring Filter)

```
= CALCULATE(SUM([Revenue]), ALL(Product_Category))
```

- What is the difference between Measure 1 and Measure 2? the difference between the measuer1 and 2 that the calculate only electronics product cat and the 2 calculate them all

EXERCISE 5: CONDITIONAL MEASURES

Objective: Create measures that apply business logic using IF and SWITCH functions.

Task:

Create conditional measures that evaluate based on business rules:

Measure 1: High Profit Orders Count

```
= COUNTX(FILTER(Sales, [Profit] > 1000), 1)
```

a) How many orders have profit greater than $1000? 229

b) What percentage of total orders is this? 45.8%

c) Filter by Region = "West". How many high profit orders in this region?

52

Measure 2: Profit Status

```
= IF([Total Profit] > 0, "Profitable",
   IF([Total Profit] = 0, "Break Even", "Loss"))
```

d) What is the overall Profit Status? profitable

Measure 3: Sales Performance Rating

```
= SWITCH(TRUE(),
    [Total Revenue] >= 500000, "Excellent",
    [Total Revenue] >= 300000, "Good",
    [Total Revenue] >= 100000, "Fair",
    "Poor")
```

g) What is the overall Sales Performance Rating? excellent

h) Filter by Region and find the rating for each region:

- North: fair
- South: fair
- East: fair
- West: fair
- Central: fair

i) Which regions are "Excellent" performers?

no one all are simillar

EXERCISE 6: TIME INTELLIGENCE MEASURES

Objective: Create measures that analyze data over time using time intelligence functions.

Task:

Create measures that perform time-based analysis:

Measure 1: Year-to-Date Revenue

```
= TOTALYTD(SUM([Revenue]), [Date])
```

a) What is the Year-to-Date (YTD) revenue for 2024? 546 K

b) Is this value larger or smaller than a monthly revenue value? Why? It is larger because it includes many months, not just one.

Measure 2: Revenue Same Period Last Year

```
= CALCULATE(SUM([Revenue]), SAMEPERIODLASTYEAR([Date]))
```

c) What was the revenue in the same period last year? 570.70K

d) Did revenue increase or decrease year-over-year? the revenu increase

Measure 3: Year-over-Year Growth

```
= DIVIDE(
    CALCULATE(SUM([Revenue])) -
    CALCULATE(SUM([Revenue]), SAMEPERIODLASTYEAR([Date])),
    CALCULATE(SUM([Revenue]), SAMEPERIODLASTYEAR([Date])),
    0
) * 100
```

e) What is the Year-over-Year growth percentage? 95.66%

f) Filter by Product Category = "Electronics". What is the YoY growth? 84.65%

g) Which product category has the highest YoY growth? home and garden

```
year_over_year_growth = DIVIDE(
   CALCULATE(SUM('Business dataset'[Revenue]))-
   CALCULATE(SUM('Business dataset'[Revenue]),SAMEPERIODLASTYEAR('Business dataset'[Date].[Date])) ,
CALCULATE(SUM('Business dataset'[Revenue]),SAMEPERIODLASTYEAR('Business dataset'[Date].[Date])),0) *100
```

CHALLENGE EXERCISES (OPTIONAL)

For advanced learners:

2. Complex CALCULATE with Multiple Filters

Create a measure comparing current selection to a specific region (North):

```
= DIVIDE(SUM([Revenue]),
    CALCULATE(SUM([Revenue]), Region = "North"), 0)
```

How does each region compare to North?

3. RANKX Function Challenge

Create a measure that ranks regions by profit:

```
= RANK([Total Profit], Table_Name)
```

Rank all regions by profit (1 = highest, 5 = lowest).

4. DISTINCTCOUNT Challenge

Create a measure counting unique customers:

```
= DISTINCTCOUNT([Customer_Name])
```

How many unique customers purchased in each region?

5. Year-over-Year Growth Analysis

Create measures comparing 2023 vs 2024 revenue with percentage growth.

- 2023 Revenue:
- 2024 Revenue:
- YoY Growth %:

SUBMISSION CHECKLIST

Before submitting your work, verify you have completed:

Exercise 1: Measures vs Calculated Columns understood with all questions answered

Exercise 2: Basic measures (Total Profit, Average Order Value, Order Count) created

Exercise 3: Calculated columns (Profit Margin, Revenue Category, Discount Flag) with Q&A

Exercise 4: Filter context measures (CALCULATE) with comprehensive analysis

Exercise 5: Conditional measures (IF, SWITCH) with business analysis

Exercise 6: Time intelligence measures (YTD, Year-over-Year) with historical analysis

Report is organized, clean, and demonstrates DAX mastery

Congratulations!

You've mastered DAX fundamentals!